

21. Orwell Bay

Shoreline length: 39 km



Shore Types

Orwell Bay is the north facing portion of the Point Prim peninsula located on the southeast end of PEI, with a majority of wetlands and cliffs.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
12	4	0	33	50	0	1	0



Cliffs and wharf remnants on north shore of Point Prim



Cliff at Belfast Highlands Greens Golf (left) and Lord Selkirk Park (right)



Cliff to wetland transition towards inner Orwell Bay

Governance, Infrastructure, and Economy

This SMU is almost completely located within the Rural Municipality of Belfast (92%). Key infrastructure includes a portion of the Trans-Canada Highway and provincial roads: Route 209 (Point Prim Road, at Eldon and Prim Point). Coastal economic drivers include the Belfast Highland Greens golf course. Local and legacy land use constraints include a few small, developed lots, several undeveloped small lots, and unpaved coastal roads.

Natural Assets and Heritage

The Lord Selkirk Provincial Park is within Orwell Bay. The shoreline includes critical habitat for bird species including the threatened Bank Swallow.

Planning and Policy Framework

Provincial rules apply. Land use and development is managed under provincial development standards, which include horizontal coastal building setbacks based on edge of feature and a calculation of erosion over time, building setbacks for dunes, and a requirement for a coastal buffer for new subdivisions. There are no specific flood elevations

identified in regulations, but coastal hazard assessments are typically conducted.

Sediment transport

The estimated net sediment transport rate averaged across the SMU is moderate. Sediment transport rate potential is highest along the north facing peninsula.

Coastal Flood and Erosion Risk

Infrastructure near the wetlands within in this area are vulnerable to increasing flood hazards. Erosion hazard is limited due to the limited amount of assets and sheltered orientation of the area up the bay.

The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential future shoreline change for areas with historical data, not including salt marshes.

Estimated number of buildings and length of road at risk [per km shoreline].

	FLOOD RISK				EROSION RISK		
	Present	Short-term 0.3m SLR (2050)	Medium-term 0.6m SLR (2080)	Long-term 1.2m SLR (2130)	Short-term (~2050)	Medium-term (~2080)	Long-term (~2130)
Buildings	0	0	0	4 [<1/km]	4 [<1/km]	4 [<1/km]	5 [<1/km]
Roads	23 m [1 m/km]	25 m [1 m/km]	106 m [3 m/km]	225 m [6 m/km]	25 m [1 m/km]	49 m [1 m/km]	104 m [3 m/km]

Proposed SMP Strategies

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
21.00 Orwell Bay	NAI	MR	MR
Rationale	Given the initially low risk levels the proposed strategy is No Active Intervention. Mid- to long-term Managed Realignment will reduce risk to the limited number of vulnerable assets while preserving natural shorelines and allowing salt marsh migration up the estuary.		

The table above shows how the recommended shoreline management unit (SMU) or policy unit (PU) strategy evolves across three planning epochs (Short-, Medium-, and Long-term) under increasing sea-level rise and erosion projections.

Adaptation strategies are preliminary recommendations subject to further development with feedback from community and government. These strategies are not approvals, commitments or policy determinations.